

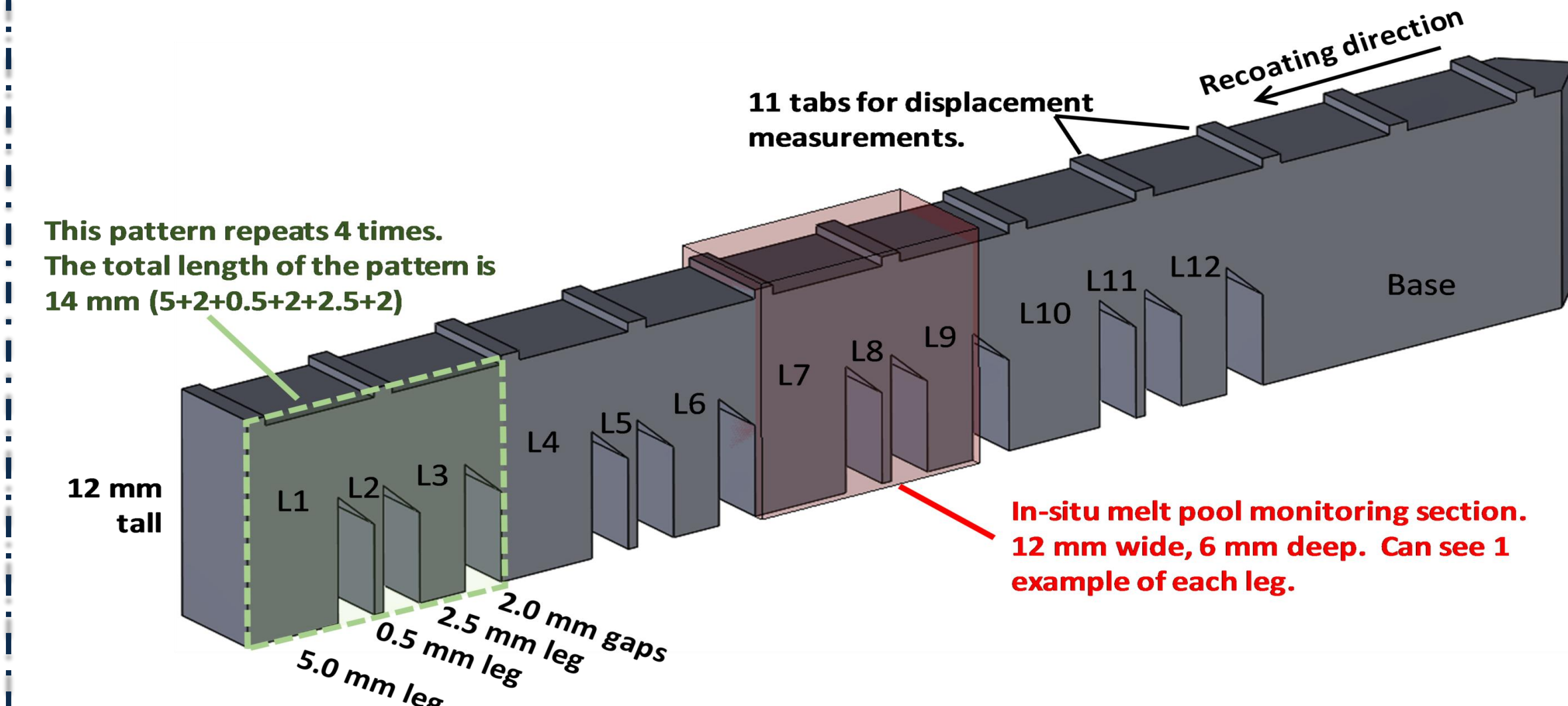
The Highlights

- Arbitrary-order sensitivities** enable the accurate characterization of nonlinear parameter interactions required for LPBF calibration, optimization, and uncertainty quantification.
- Large-scale LPBF simulations require scalable CPU and GPU** parallelism to manage the computational cost of fine discretizations, nonlinear physics, and sensitivity analysis.
- Kokkos/HYPAD** provides performance-portable arbitrary-order automatic differentiation across CPU and GPU architectures.
- The Residual/HYPAD method** computes responses and arbitrary-order sensitivities simultaneously in a single nonlinear simulation.
- New thermo-mechanical formulations** capture heat transfer, phase transformations, moving heat sources, and residual stress development beyond MFEM's native capabilities.



1. Benchmark (NIST AMB 2018-01)

1.1. Problem setup

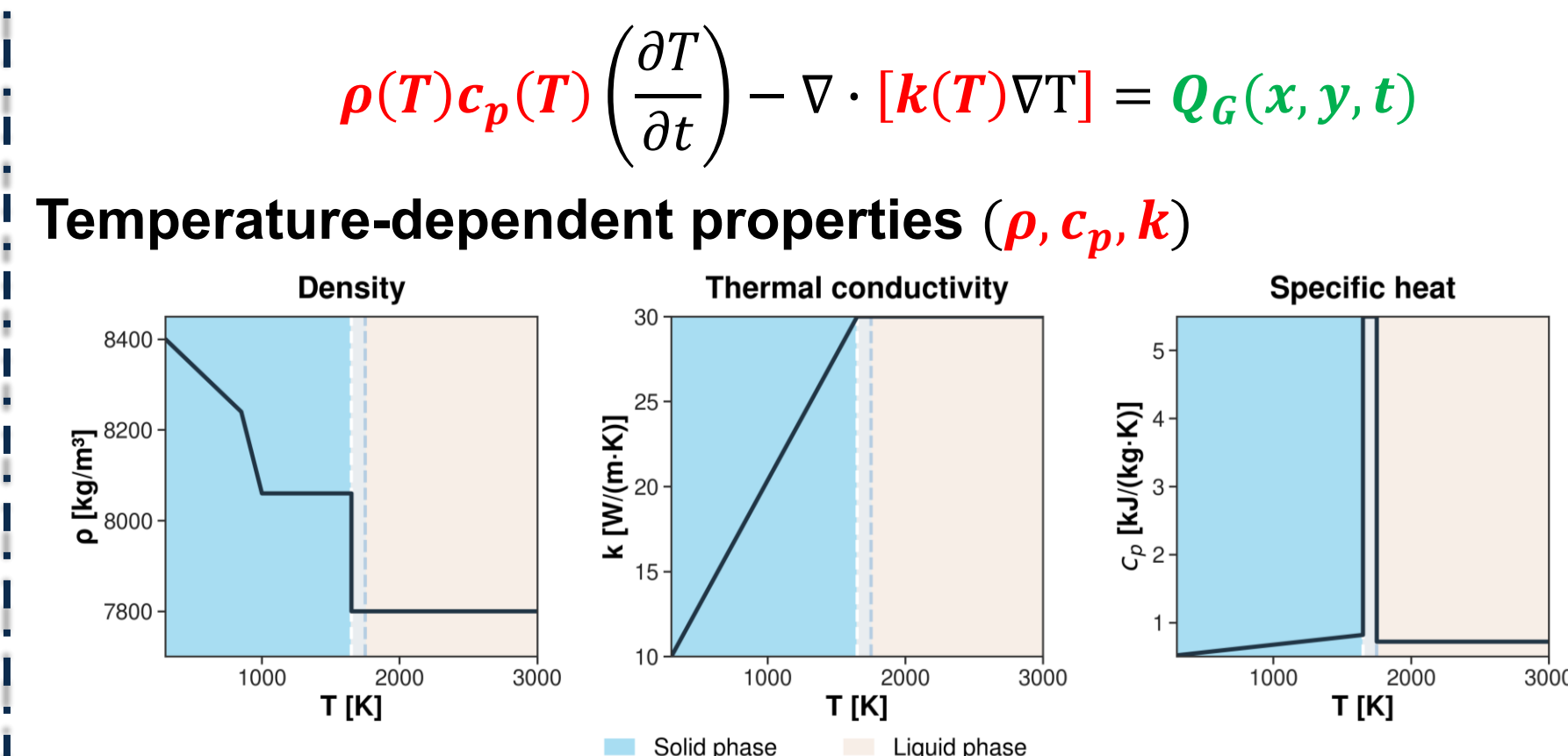


- Element count:** 278,834 with 82,010 in the substrate and 196,824 across 4 layers.
- Element type:** linear hexahedra (8-node trilinear brick, MFEM geometry type 5).
- Material:** nickel-based superalloy IN625, and 15-5PH stainless steel.

Physics	Parameter (a_i)	Symbol	Value	Units
Laser	Radius	(r_x, r_y, r_z)	(0.1, 0.1, 0.1)	mm
	Absorption	η	0.43	—
	Power	P	195	W
	Initial Location	(x_0, y_0, z_0)	(0, 0, 0)	mm
	Scan Speed	(V_x, V_y, V_z)	(0.8, 0, 0)	m/s
Boundary condition	Convection	h	18	$W/m^2 \cdot K$
	Build chamber temperature	T_∞	303.15	K
	Emissivity	ϵ	0.4	—
Initial condition	Initial temperature	T_0	303.15	K
	Energy	L	290	kJ/kg
Phase change	Solidus temperature	T_1	1563.15	K
	Liquidus temperature	T_2	1623.15	K
	Temperature density	$\rho(T)$	T dependent	kg/m^3
Material properties	Thermal conductivity	$k(T)$	T dependent	$W/m \cdot K$
	Specific Heat	$c_p(T)$	T dependent	$kJ/kg \cdot K$

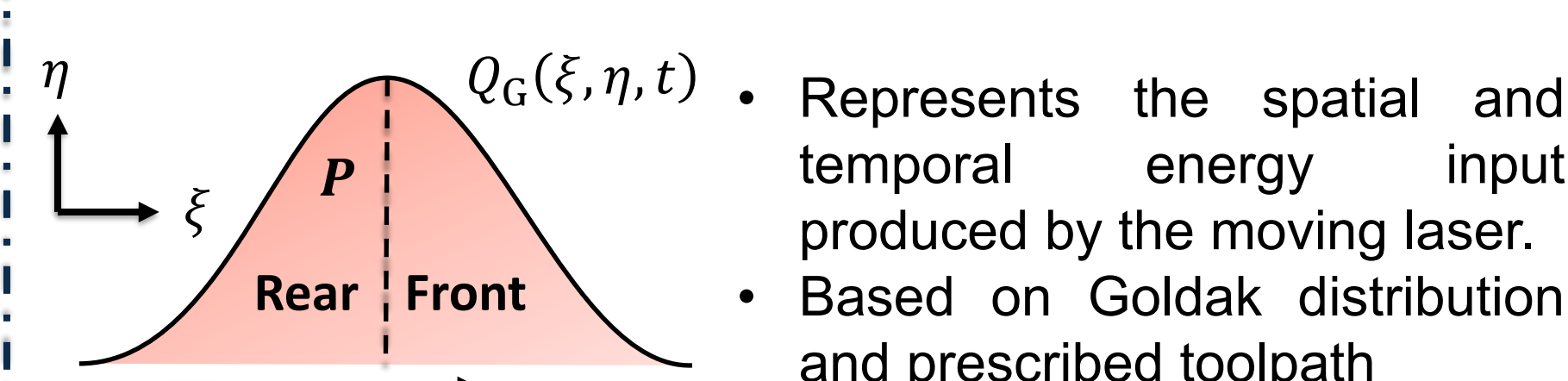
1.2. New physics added to MFEM

1.2.1 Transient nonlinear heat conduction



Update properties according to the current temperature based on the external file loaded

Goldak moving heat source ($Q_G(x, y, t)$)



- Represents the spatial and temporal energy input produced by the moving laser.
- Based on Goldak distribution and prescribed toolpath

1.2.3 MFEM thermal integrators

$$\underbrace{M(T^{n+1}) \frac{T^{n+1} - T^n}{\Delta t}}_{\text{Temperature-dependent capacitance}} + \underbrace{K(T^{n+1}) T^{n+1}}_{\text{Temperature-dependent conductivity}} + \underbrace{F_{\text{Goldak}}(t^{n+1})}_{\text{Moving heat source}} = 0$$

- Thermal capacity integrator $M(T)$:** governs the rate of heating/cooling and the latent heat non-linearity (melt/solidification).
- Thermal diffusion integrator $K(T)$:** governs how heat spreads away the melt pool into the surrounding solid/powder bed.
- Goldak heat-source integrator $F_{\text{Goldak}}(t)$:** applies the moving volumetric laser heat input based on the Goldak distribution and the prescribed toolpath

1.2.4 MFEM mechanical integrator

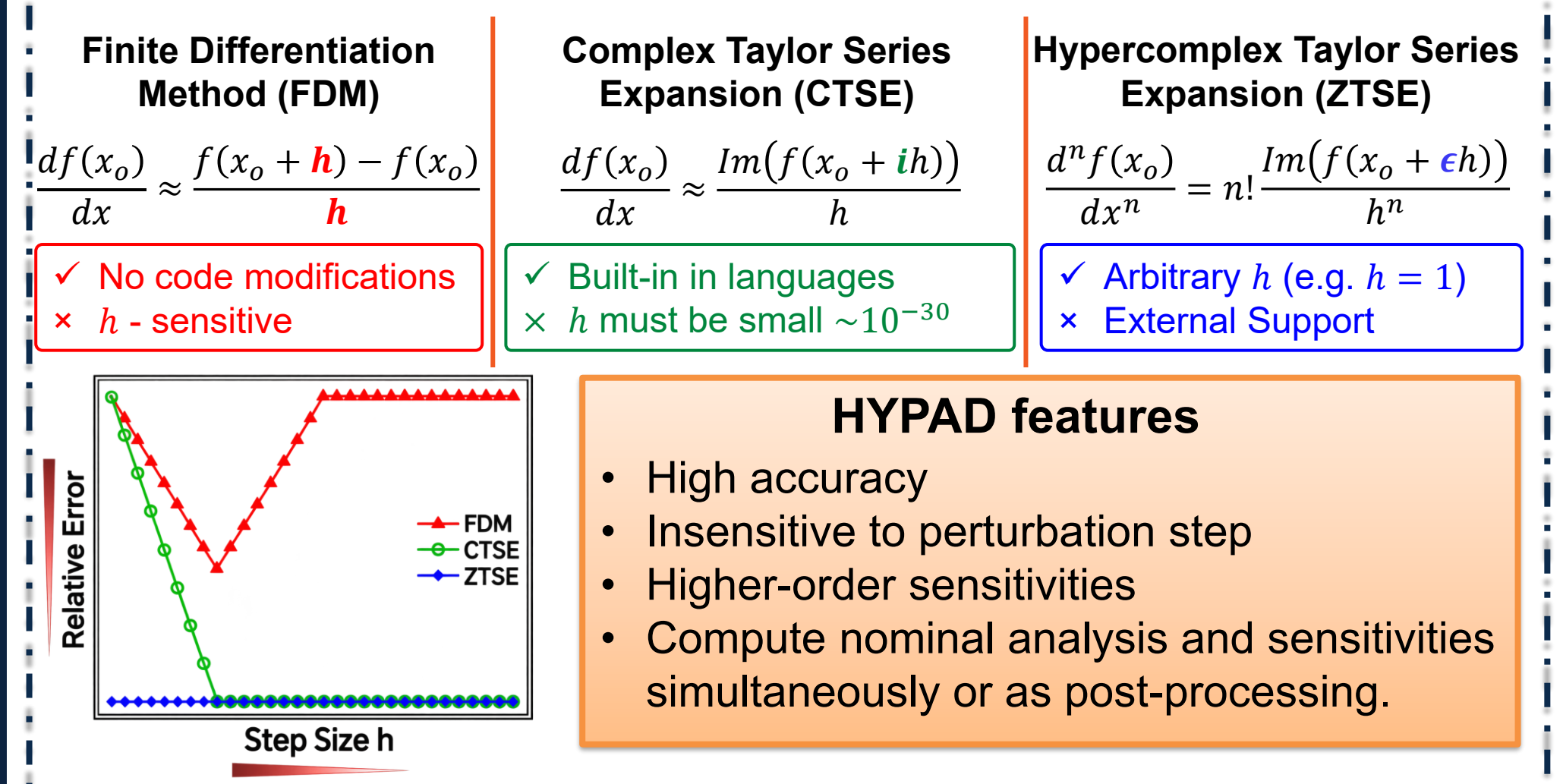
Hill anisotropic plasticity (Φ_{Hill})

Extends the von Mises criterion to direction-dependent yielding using six Hill coefficients: F, G, H , derived from normal directional yield stresses (σ_i), and L, M, N , derived from shear yield stresses (τ_{ij}).

$$\Phi_{\text{Hill}} = \sqrt{F(\sigma_{22} - \sigma_{33})^2 + G(\sigma_{33} - \sigma_{11})^2 + H(\sigma_{11} - \sigma_{22})^2 + 2L\sigma_{23}^2 + 2M\sigma_{13}^2 + 2N\sigma_{12}^2 - \sigma_y} \leq 0$$

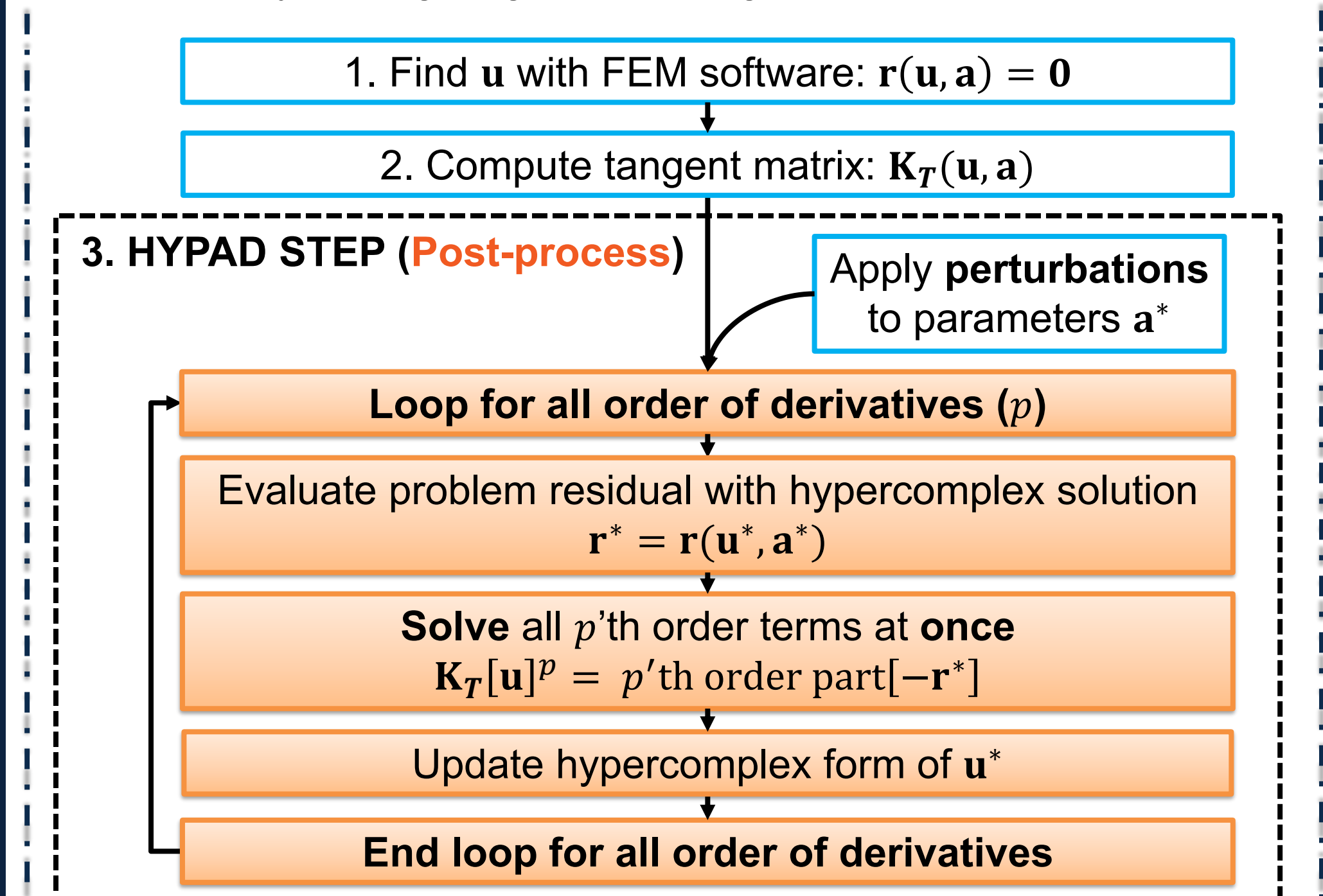
2. Key Concepts

1.1. HYPAD derivative workflow



1.2. Residual/HYPAD method

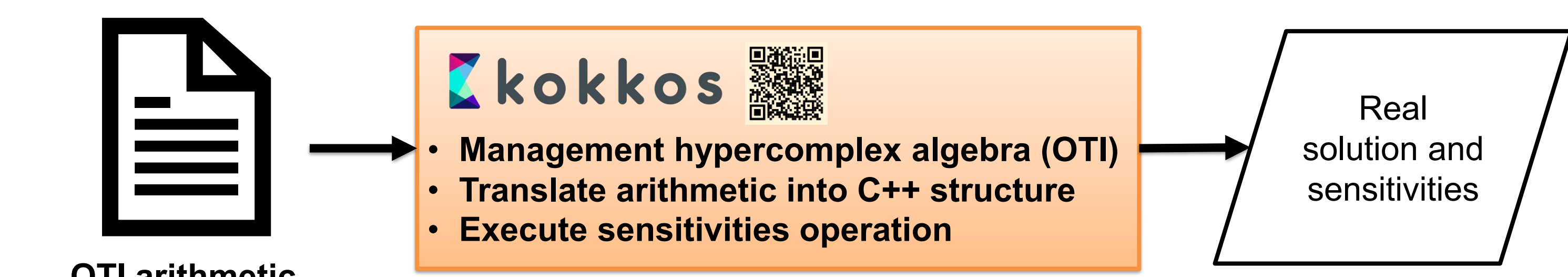
HYPAD embeds hypercomplex algebra (OTI) directly in the residual formulation $r(u, a) = 0$. An enriched solver returns a real solution and sensitivities for any user-defined parameters, automatically propagating order through elemental definitions.



For more details: Rincon-Tabares, J.S., Aristizabal, M., Balcer, M., Montoya, A., Millwater, H. and Restrepo, D., 2024. Efficient Sensitivity Analysis of the Thermal Profile in Powder Bed Fusion of Metals using Hypercomplex Automatic Differentiation Finite Element Method. Additive Manufacturing.

3. Kokkos portable type

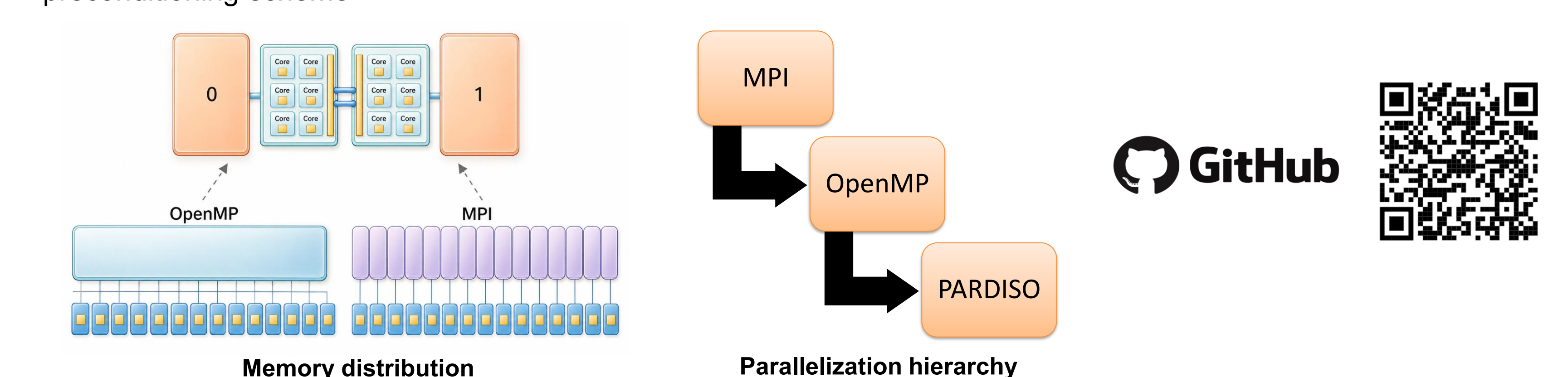
- Device-portable OTI type:** a single portability macro selects Kokkos or standard C++ storage, letting the same OTI arithmetic run on CPUs for any parameter and derivative order
- Portability:** ensures the method is usable on any workstation – no GPU allocation or specialized software.



For more details: Avans, C. N., Dos Santos, G., Chevalier, C., Taboada, H., Pérache, M., Dosanjh, M. G. F., Olivier, S. L., Pearson, C., Suggs, E. D., Kale, V., & Skjellum, A. (2026). Kokkos Comm: Performance Portable Communication for Distributed Kokkos Applications. *Proceedings of the Platform for Advanced Scientific Computing Conference*

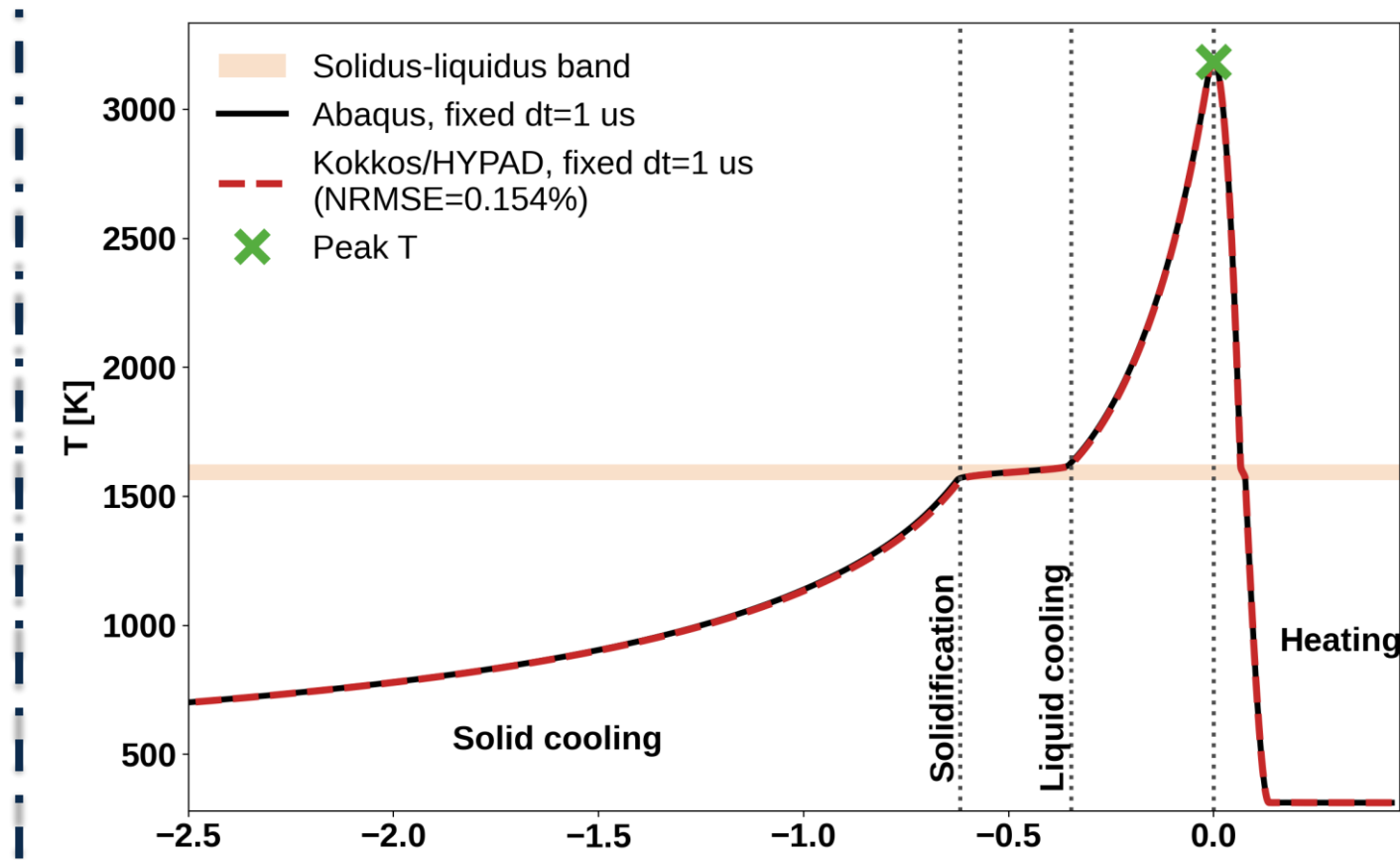
4. Parallelization layer

- Multi right-hand-side PARDISO wrapper:** packs all sensitivity directions into one phase call.
- OpenMP (shared memory, separate process):** parallelize the element loop over threads and cache quadrature-point geometry data.
- MPI (distributed memory, separate process):** domain rank-based decomposition solver with a nested PARDISO preconditioning scheme



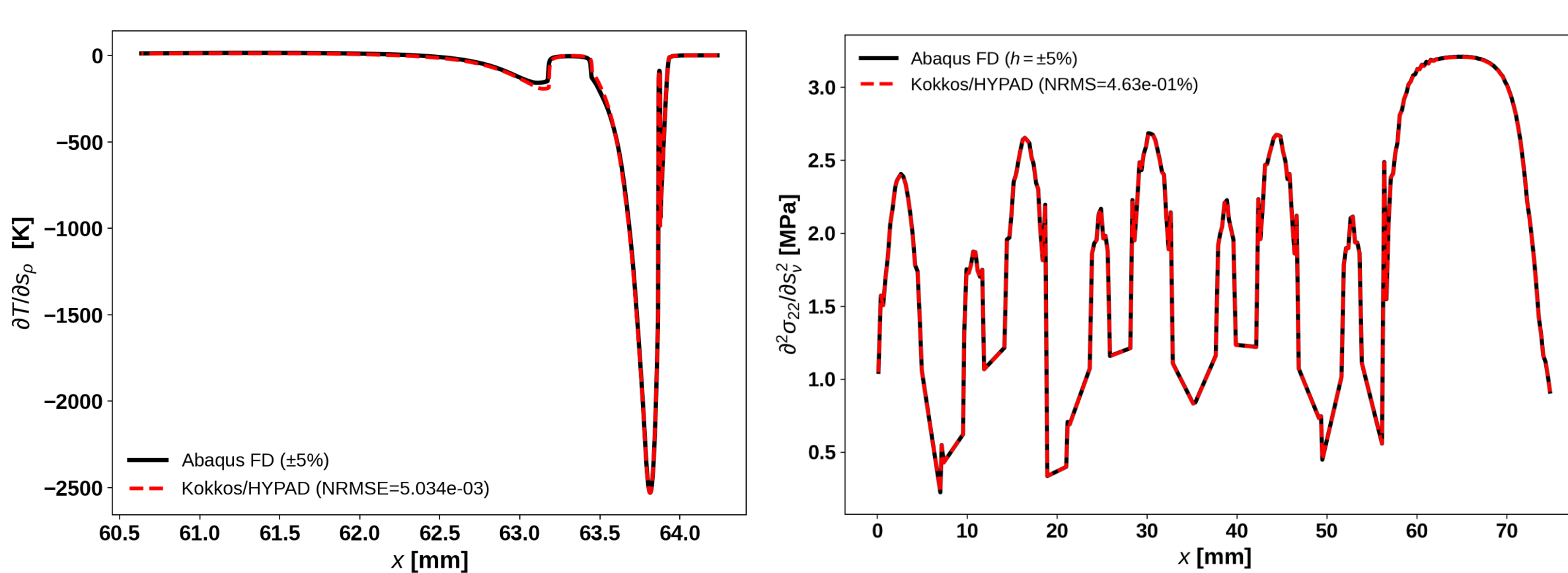
5. Results

4.1. Nominal solution



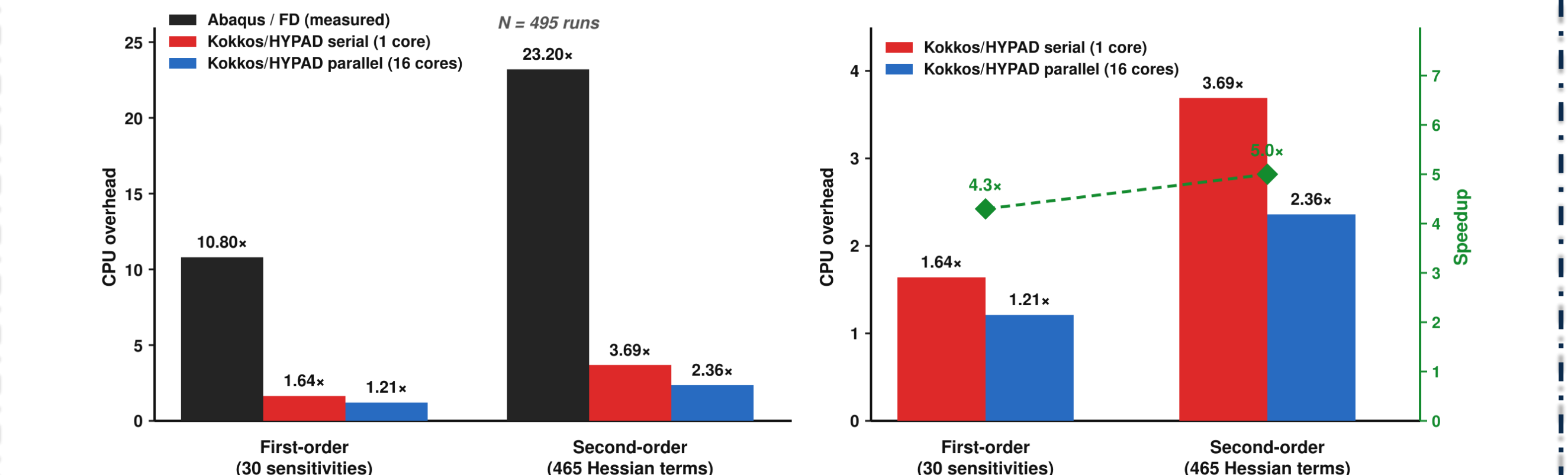
MFEM-Kokkos-HYPAD reproduces Abaqus temperature profile (NRMSE $\leq 0.2\%$).

4.2. Mechanical and thermal derivatives



Excellent agreement on thermal and mechanical sensitivities (NRMSE $\leq 1\%$) First- and second-order

4.3. Computational cost



495 Abaqus runs required for all FD 465 second-order Hessian terms (30 parameters) Requires 1 nominal baseline, 30 single-parameter perturbation runs, 434 mixed-pair runs, and 30 additional runs for the pure diagonal second derivatives

5. Conclusions

- A transient nonlinear thermal FEM framework for LPBF is developed with temperature-dependent properties, latent heat, and moving heat-source modeling.
- Kokkos/HYPAD reproduces Abaqus thermo-mechanical results for the NIST AM-Bench 2018-01 benchmark with NRMSE below 0.2%.
- First- and second-order sensitivities computed with Kokkos/HYPAD closely match Abaqus finite-difference derivatives, with NRMSE values between 0.006% and 0.5%.
- Thermo-mechanical operations are parallelized across multicore CPUs, while Kokkos manages workload distribution and thread-level execution.
- Kokkos/HYPAD delivers greater accuracy and computational efficiency than commercial finite-difference-based thermo-mechanical AM solvers.

Resources



Support

